

SKY HIGH (TOWER MAKING) CHALLENGE

Event Description:

The SKY HIGH is a Tower Making challenge for participants to design and construct the tallest and most stable tower using the materials provided within a fixed time limit. This event tests participants' creativity, structural understanding, problem-solving ability, and teamwork. The objective is to encourage practical application of engineering principles while promoting collaboration and healthy competition.

Eligibility Criteria:

- Open to students from all departments and colleges.
- Participation is team-based only.
- Each team must consist minimum of 2 students and a maximum of 5 students.
- Prior registration is mandatory.
- Basic understanding of structural concepts is recommended but not mandatory.

Rules & Regulations:

1. Participation Guidelines

- Teams must follow the given size, capacity, or format limits.
- Only the materials provided by the organizers may be used (no personal materials allowed).
- Prohibited items or practices (list based on event type) are not allowed.
- All participants must follow safety and discipline protocol.

2. Match/Performance Rules

- The competition will be conducted in only one round.
- Each team will be given a time limit of 2 hours to complete their tower.
- Elimination occurs if:
 - Towers fail to meet the minimum height requirement (50 cm).
 - Teams exceed the 2-hour time limit.
 - Teams violate safety or performance rules.

3. Evaluation / Scoring Criteria

1. Minimum Height Requirement:
 - Towers must have a minimum height of 50 cm to qualify for evaluation.
 - Towers below 50 cm will not be considered for scoring.
2. Stability:
 - The tower must remain freestanding and stable for at least 30 seconds after construction is complete.

- Wobbling or partial collapse during this time will reduce points.
 - Load-Bearing Capacity: After passing stability, a load test will be conducted by gradually placing weights on the tower. Higher load-bearing capacity earns higher scores.
3. Earthquake Resistance
- Place the tower on a vibration platform or a flat surface that can simulate shaking.
 - Apply vibrations at a certain frequency, gradually increasing the frequency or amplitude.
 - Observe how the tower withstands the vibrations:
 - Slight wobble but stands → Good
 - Moderate wobble, minor displacement → Fair
 - Collapses or topples → Poor
 - Scoring: Towers that survive higher frequencies without collapsing get higher points.